1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L12 – Interface Design Guidelines I**

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4 **Recap**

- Earlier, we got a general introduction to design
- Two concerns
  - Interface (& interaction) design
  - Code design

5 **Recap**

- Today, we shall start with interface design – where to start
- We shall learn about design guidelines

6 **Design Guidelines**

7 **Recap**

- Provide starting point in the development life cycle

8 **Guideline Types**

- Many guidelines are there

9 **Generic Guidelines**

- Some very generic and consequently small in size - covers broad aspects of interactive systems, at a rather high level
  - "Eight golden rules" (Shneiderman, 1986)
  - "Seven principles" (Norman, 1988)

10 **Specific Guidelines**

- Others are more detailed and specific and therefore large in (set) size - intended to cover minute aspects of the design, often for specific products
  - "human interface guidelines" for the Apple systems

11 **Shneiderman's Guidelines**

12 **Golden Rules (Shneiderman, 1986)**

- A set of eight rules
- Originally designed for GUIs
- Partly (or fully) applicable for other types of interfaces as well

13 **Rule 1**

- Strive for consistency
  - Internal consistency
  - External (environmental) consistency

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15 **Rule 2**

- Design for universal usability
  - For novice, intermittent, expert (e.g., menu and hotkey)

16 **Rule 3**

- Offer informative feedback
  - E.g. progress bars
  - Color change of the floppy (storage metaphor) after save operation is done

17 **Rule 4**

- Design dialogues to yield closure
  - Related to previous rule
  - Organize activities into groups– beginning, middle, end
  - Some feedback at the end of each group
  - E.g. – online shopping (lots of subtasks, grouping helps)

18 **Rule 5**

- Offer error prevention and simple error handling

- Design to keep error rates low (close and start options should not be kept close to each other)
- Don't show complicated error message

19 **Rule 6**

- Permit easy reversal of actions
  - Undo and redo

20 **Rule 7**

- Keep users in control
  - Let the user feel that they are in control
  - User should be able to perceive their interaction and change in system state (e.g., drag and drop)

21 **Rule 8**

- Reduce short-term memory load
  - Gorge A Miller (1956) – "7±2" rule
  - Should not force users to remember too many things

22 **Case Study**

23 **The Case**

- A web page for railway ticket reservation - a traveler wants to book tickets online
  - Note - traveler likely to be non-(technology) expert
  - Therefore, usability concerns very relevant

24 **A Solution**

25 **Problem**

- In order to understand shortcomings of proposed solution (interface) w.r.t. ticket booking task, let us first analyze the task from a user's perspective

26 **The (Ticket Booking) Task**

- The task actually involves a series of sub tasks to be performed in the following sequence

27 **The (Ticket Booking) Task**

- User enters source and destination station details to know about all the trains running between the stations, along with their timings, fare and seat availability information

- Based on the train information, the user selects one train
- Makes online payment to book the ticket

28 **The (Ticket Booking) Task**

- Usability consideration in each of these sub tasks is very important
- For simplicity, let us consider only the first sub task: providing the source and destination information
- How can a user do that with the interface?

29 **The (Ticket Booking) Task**

30 **The (Ticket Booking) Task**

31 **The (Ticket Booking) Task**

32 **The (Ticket Booking) Task**

- Unfortunately, the hyperlink only allows user to search for trains between "important" pair of stations
- For users travelling from a so-called "unimportant" station or to an "unimportant" station or have both source and destination stations categorized as "unimportant", it is of no use

33 **The (Ticket Booking) Task**

- Such users are likely to constitute a significant population (since number of "important" stations is much less compared to the total number of stations in a railway network)

34 **The (Ticket Booking) Task**

- Indeed, this simple knowledge did not escape the attention of the development team and there is in fact another hyperlink to search for trains between "any pair" of stations
- Can you spot the hyperlink?

35 **The (Ticket Booking) Task**

36 **The (Ticket Booking) Task**

37 **The (Ticket Booking) Task**

38 **The (Ticket Booking) Task**

39 **The (Ticket Booking) Task**

40 **The (Ticket Booking) Task**

41 **The (Ticket Booking) Task**

- Let's summarize the activities a user is supposed to perform to be able to complete "just" the first step of the ticket-booking task: learn about the trains running between a pair of stations

42 **The (Ticket Booking) Task**

- Locate the appropriate hyperlink on the screen
- Enter station codes for source and destinations

- Use the hyperlink provided to learn about the station codes, if not known already

43 **Design Revision using Guidelines**

- Intermittent users not likely to remember station codes
- If forced to use another hyperlink to learn about station code, might create inconveniences

44 **Design Revision using Guidelines**

- Such designs in turn might violate some of the eight golden rules of interface design by Shneiderman (in particular, the 2nd rule on the "design for universal usability", the 7th rule on "keep users in control" and the 8th rule on "reduce short term memory load")

45 **Design Revision using Guidelines**

- To conform to these guidelines, it is preferable user gets help on station codes in the same interface itself, rather than requiring to switch between interfaces

46 **Another Solution**

- One way to do that is to use a "predictive" text input field

47 **Another Solution**

- User enters characters for station name
- As soon as user starts entering characters, system "predicts" station name and code
- Predictions shown on a drop-down list

48 **Another Solution**

49 **Another Solution**

- If user finds station name matching, s/he simply selects the code from the list

50 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

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